

Introduction

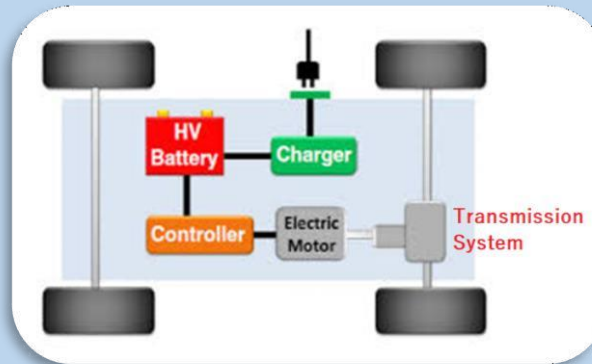
- Urban areas face challenges with high traffic emissions, causing air pollution, noise, and environmental degradation.
- Traditional fossil-fuel vehicles are labor-intensive to maintain, costly to run, and inefficient for daily urban transport.
- Electric cars collect, transfer, and utilize clean battery power, reducing fossil fuel dependence and improving efficiency.
- Electrified transportation is easier to operate and can be powered by renewable energy, supporting sustainable mobility.
- Transitioning to electric vehicles helps cities maintain better environmental standards while lowering overall commuting effort.

Purposes

- To reduce dependence on fossil-fuel vehicles
- To promote clean transportation
- To study modern technology of Electric Vehicles (EVs)
- To design for small scale transportation
- To fabricate a typical Electric Vehicle (EV)

Working Principle

- Electric vehicle work on the principle of transforming electric energy into mechanical energy, EVs feature battery and electric motor instead of a conventional fuel engine.



Specification and Size

Technical Category	Dimension (SI Unit)
Length of the vehicle	2.794 m
Width of the vehicle	1.016 m
Height of the vehicle	1.524 m
Vehicle weight + Four Passengers' weight	470 kg
Vehicle max: speed	30 km/hr
Travel distance (Fully charged)	20 km
No. of Batteries (12 V, 20.3 Ah) Lead Acid	5 Nos.
Motor power	3 kW
Wheel diameter	0.381 m
No. of drum brakes	2
No. of leaf Springs	2



**Technological University
(Mandalay)**

Department of Mechanical Engineering

“Electric Vehicle (EV Car)”



A Quiter Road to Tomorrow

A quiet hum upon the road,
A lighter step, a gentler load.
The air is clear, the sky is wide,
With a quiet future as our guide.



Fabrication



General Components



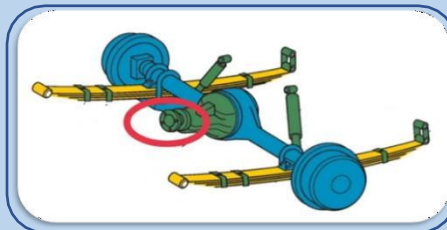
(1) Frame and tyres



(2) Steering System



(3) Braking System



(4) Transmission and Suspension System

